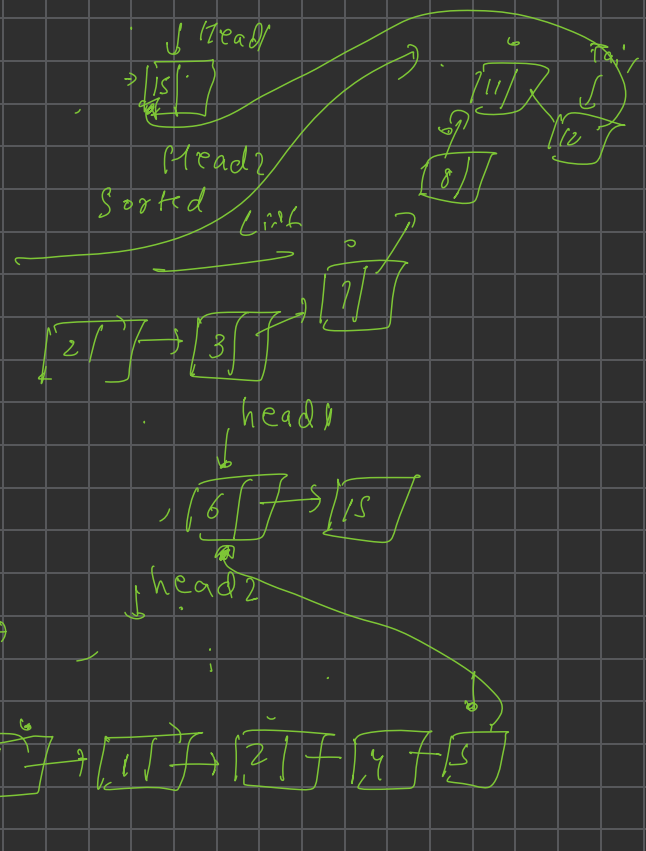
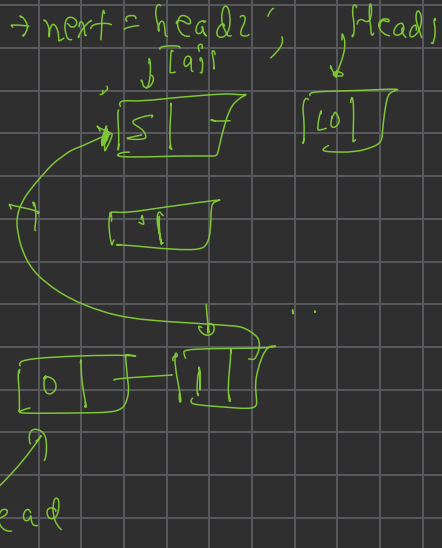
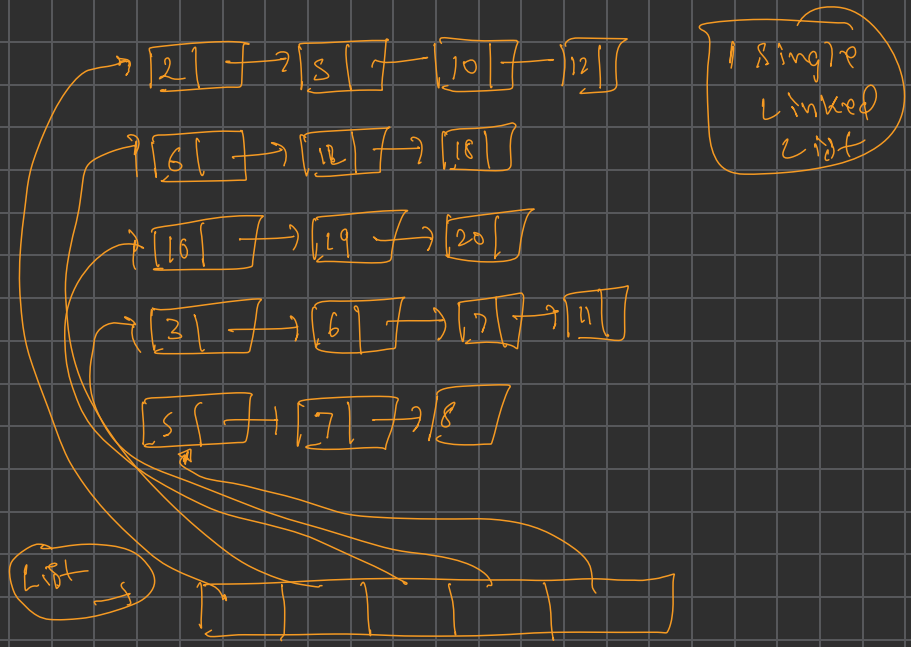
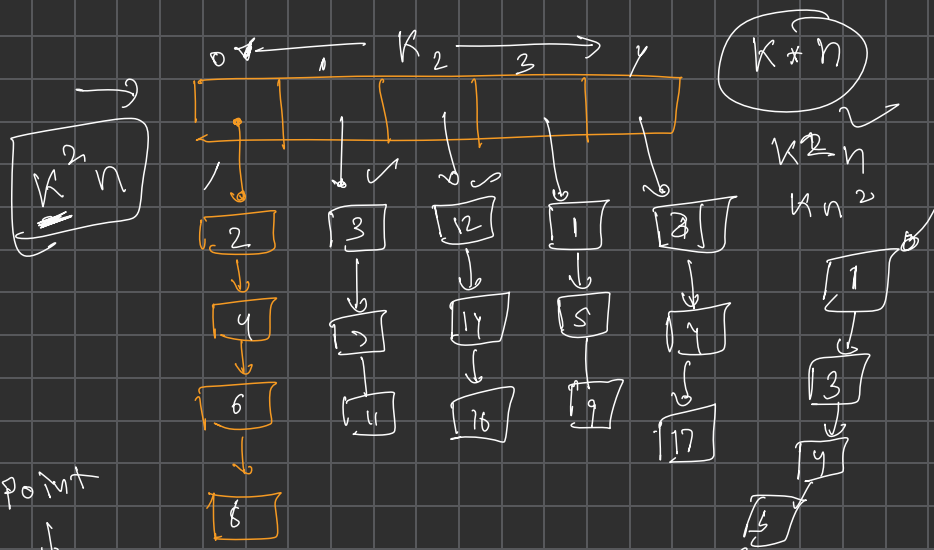


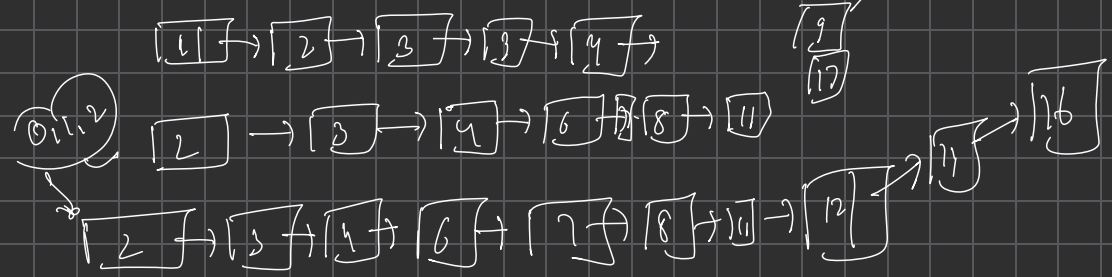

```
Node * Head = new Node(0)
    Node * Tail = Head;
while (head1 && head2)
{   if (head1 -> data <= head2 -> data)
    {   Tail -> next = head1
```

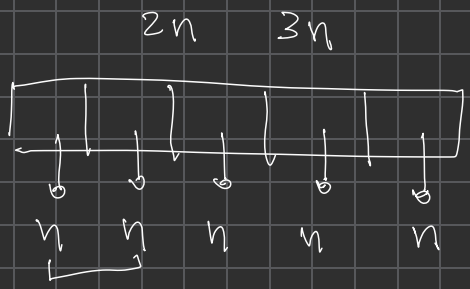
```
            head1 = head1 -> next;
            Tail = Tail -> next;
            Tail -> next = NULL;
        }
    else
    {
            Tail -> next = head2;
            head2 = head2 -> next;
            Tail = Tail -> next;
            Tail -> next = NULL;
        }
    }

if ( ! head2 )
    Tail -> next = head1;
else
    Tail -> next = head2;
```




2 → 5 → 10 → 12
6 → 12 → 18
10 → 19 → 20
3 → 6 → 7 → 11
5 → 7 → 8
List
single linked list


0 1 K 2 3 4
K*n
$K^2 n$
$K^2 n$
Kn^2
2 4 6 8
3 5 11
12 14 16
1 5 9
3 7 17
1 3 4 5 9 12


point
1 → 2 → 3 → 4 → 5 →
0,1,2
2 → 3 → 4 → 6 → 7 → 8 → 11
2 → 3 → 4 → 6 → 7 → 8 → 11 → 12 → 14 → 16



add karna hai

$$(2n + 3n + 4n + 5n)$$

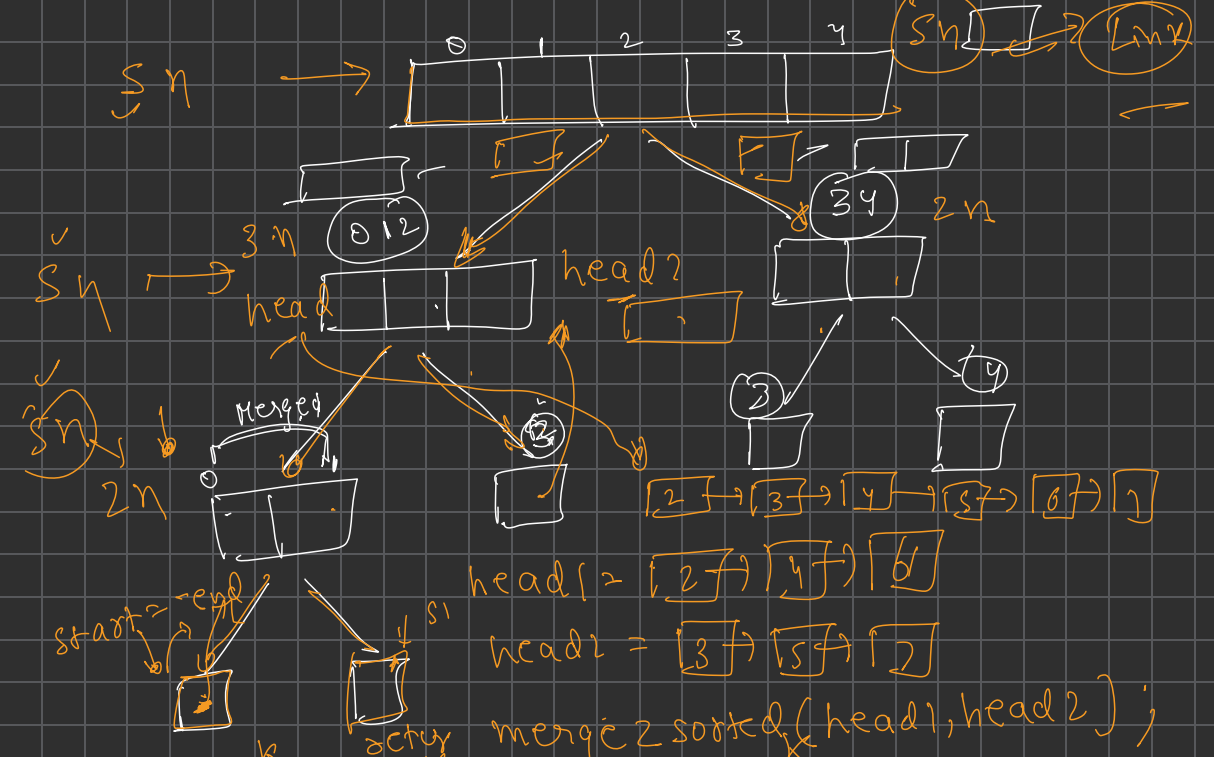
$$(2n + 3n + 4n + \cdots kn)$$

$$n(2 + 3 + 4 + \cdots k)$$

$$n\left(k \times (k+1)/2\right)$$

$$\boxed{k^2 n} \Rightarrow \text{Time complexity}$$

simply



head1 = 2 → 4 → 6

head2 = 3 → 5 → 7

merge2sorted(head1, head2);

$S_n + S_n + \dots n + S_n$

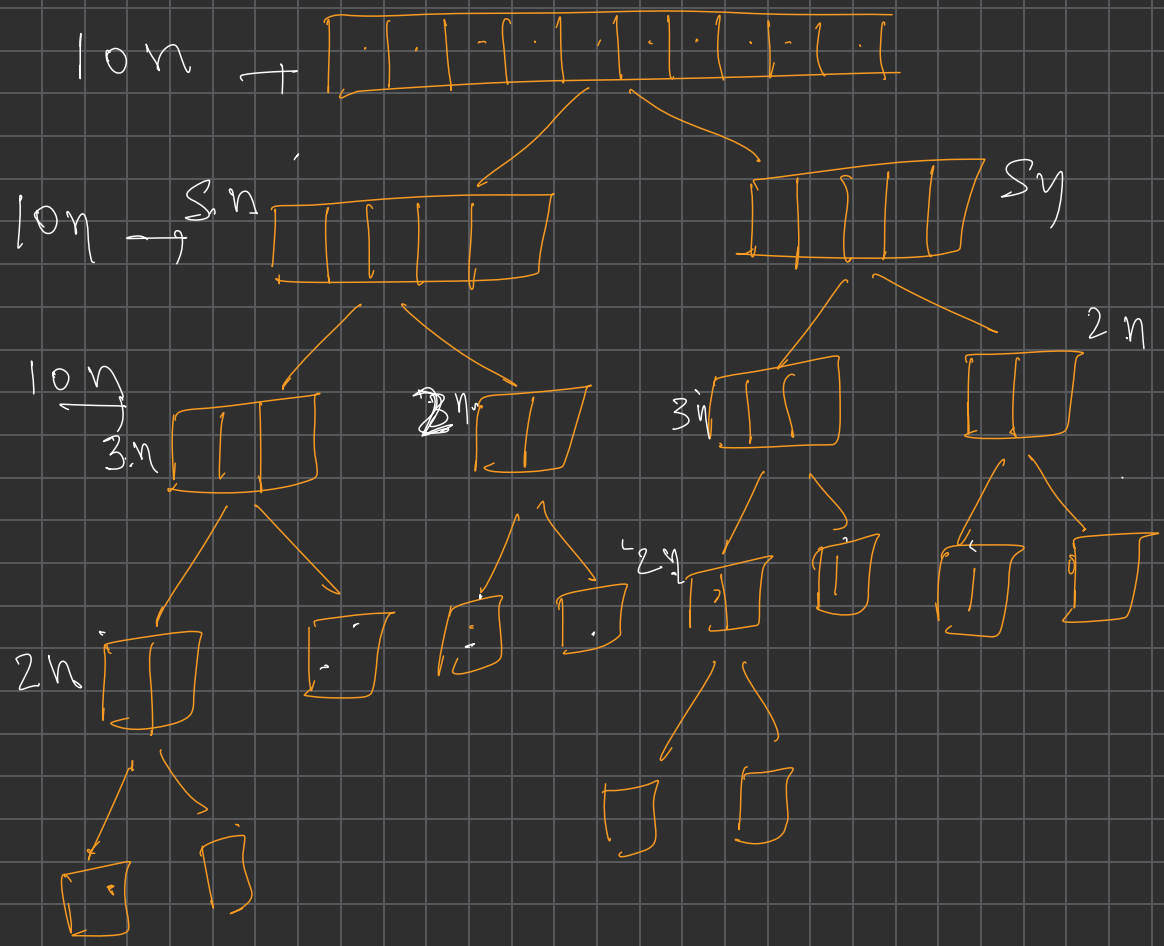
$(Kn + Kn + Kn + Kn)$

$\log K * Kn$

$K \log K\, n$ = time

$K^2 n$

$K \log K\, n$

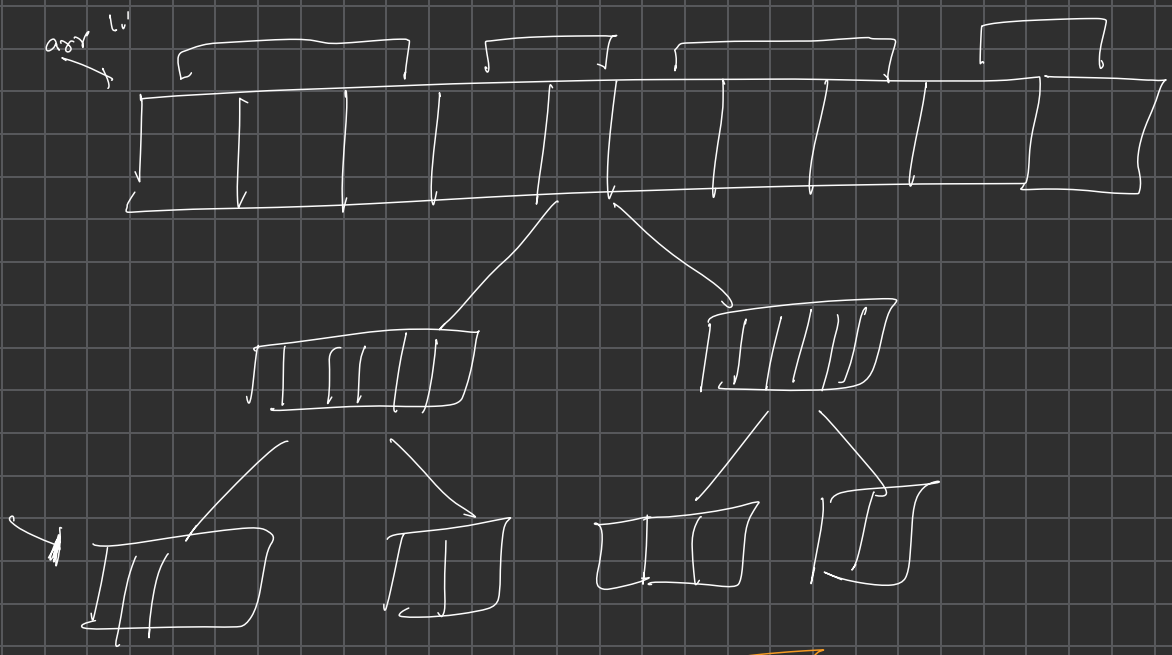


$$10n + 10n + 10n + 10n$$

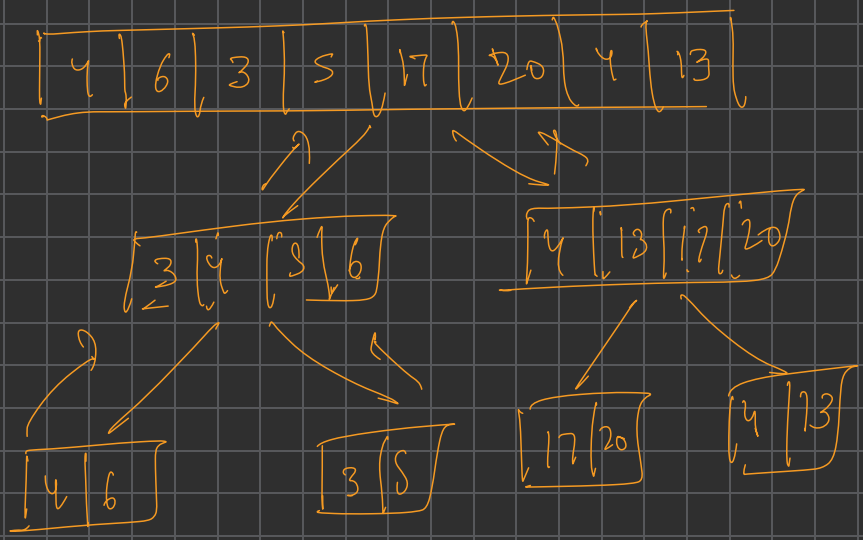
$$[Kn + Kn + Kn + \dots]$$

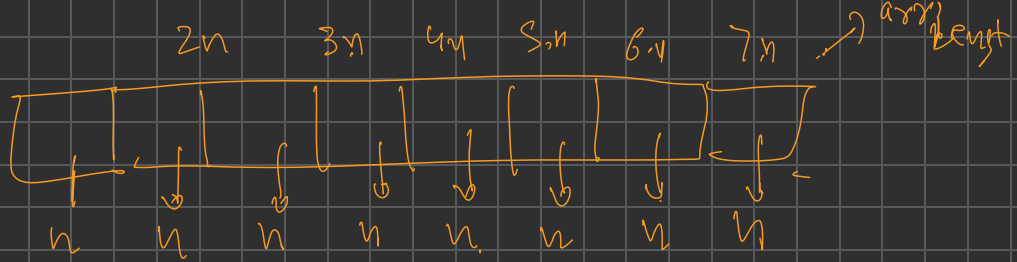
$$\log_K Kn$$

$$\boxed{K \log_K n}$$

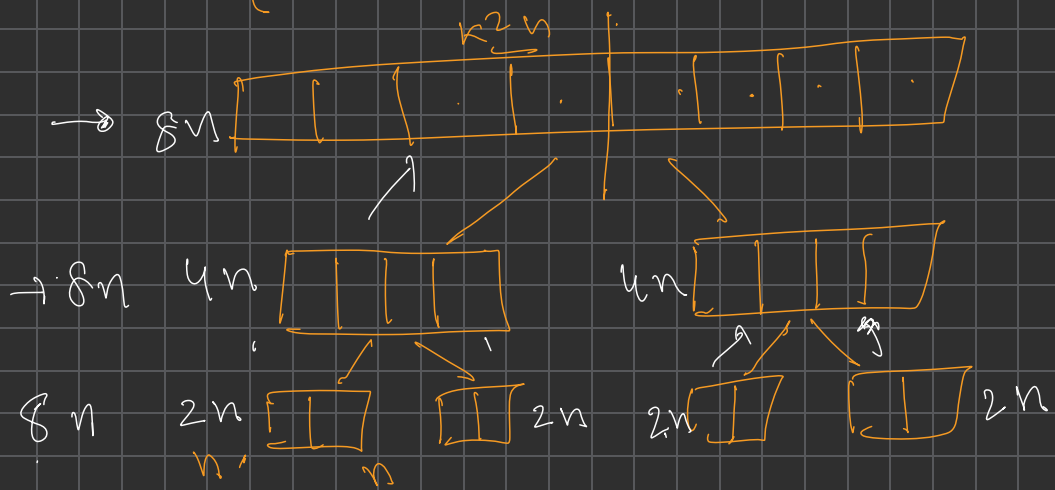


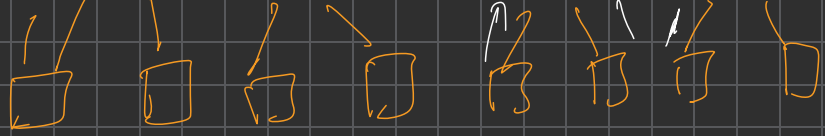

```
Node*
{
    merge(vector<Node*>&arr, start, end)
    if(start == end)
        return arr[start];

    int mid = start + (end - start)/2

Node *head1 = merge(arr, start, mid)
Node *head2 = merge(arr, mid+1, end);
return merge2sortedList(head1, head2)

}
```



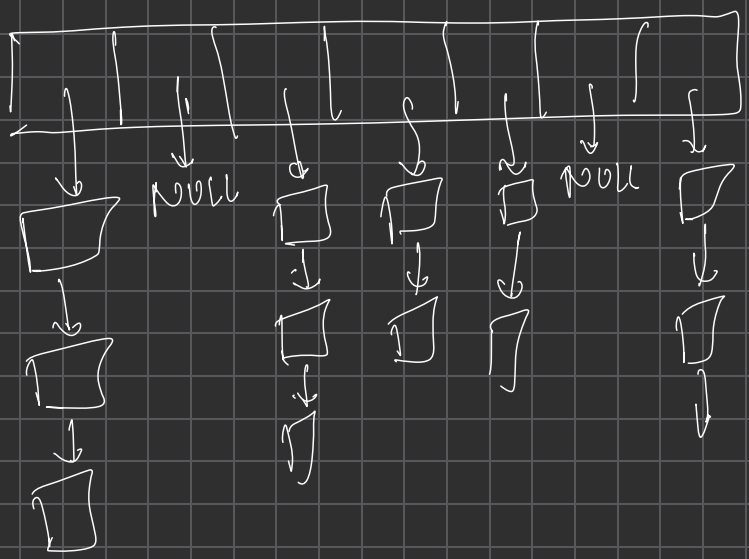


$$\frac{(2n+3n+4n+5n+6n+7n+8n)}{n(2+3+4+5+6+7+\cdots K)}$$



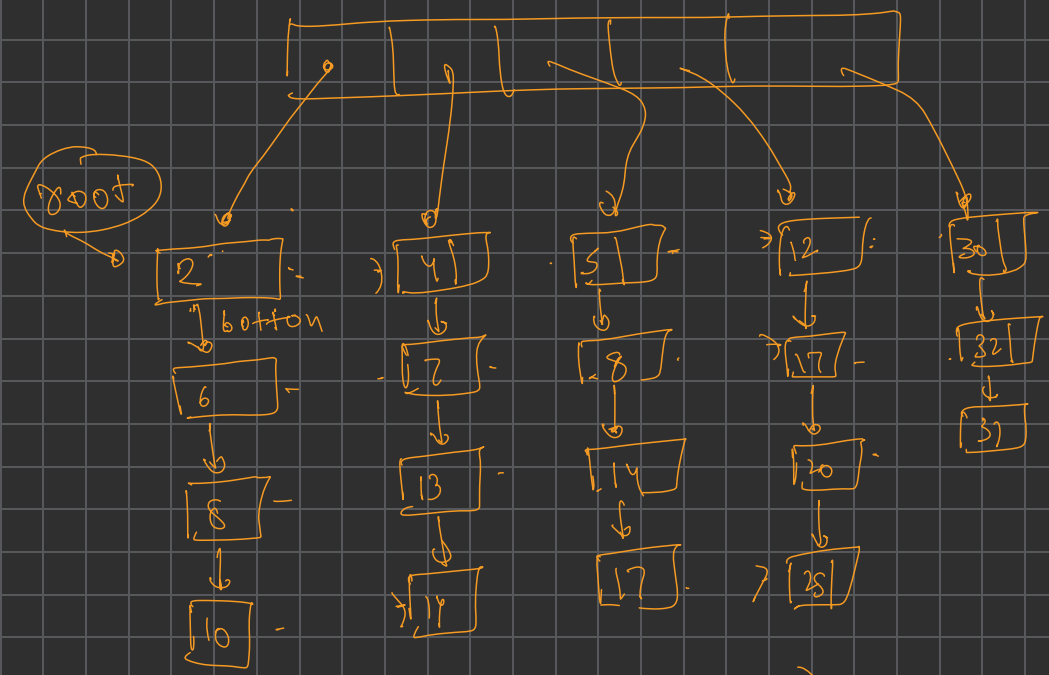


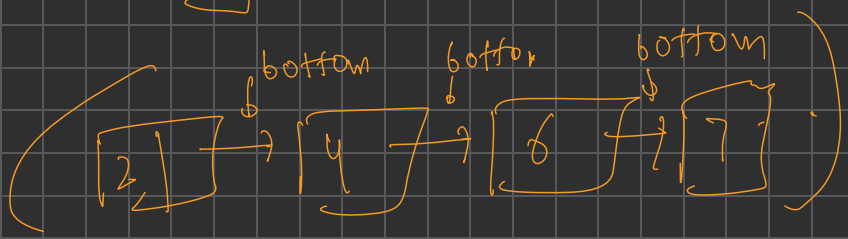
$\boxed{\log kn}$



head1 = ✓
head2 = null

val | nex | bottom

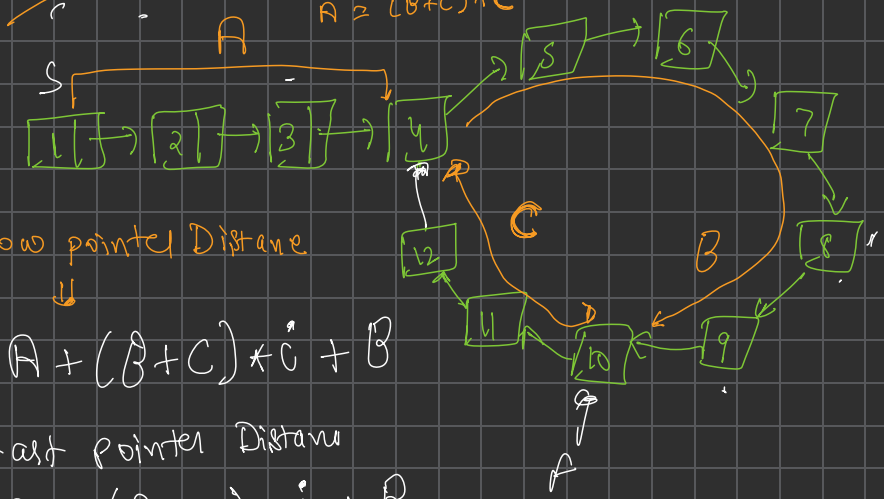

root
2
4
5
12
30
bottom
6
7
8
17
32
8
13
14
20
31
10
14
17
25


bottom
bottom
bottom
2
4
6
7

$j > 2i$

i = Total chakker slow (no of round

$A = (B+C)*C$



(1) Slow sin
2 fast ste
Once they mp
Slow = Fcal
dono ek

Slow pointer Distance

$\Downarrow$

(1) $A + (B+C)*i + B$

Fast pointer Distance

(2) $A + (B+C)*j + B$

[Fast = 2*slow]

$A + (B+C)*j + B = 2A + 2(B+C)*i + 2B$

$A + B = (B+C)*j - 2(B+C)*i$

$A + B = (B+C)*(j - 2i)$

$\boxed{j > 2i}$

$A + B = X(B+C)$

$X =$

$A + B = 2B + 2C$

$A + \cancel{B} = \cancel{B} + C$

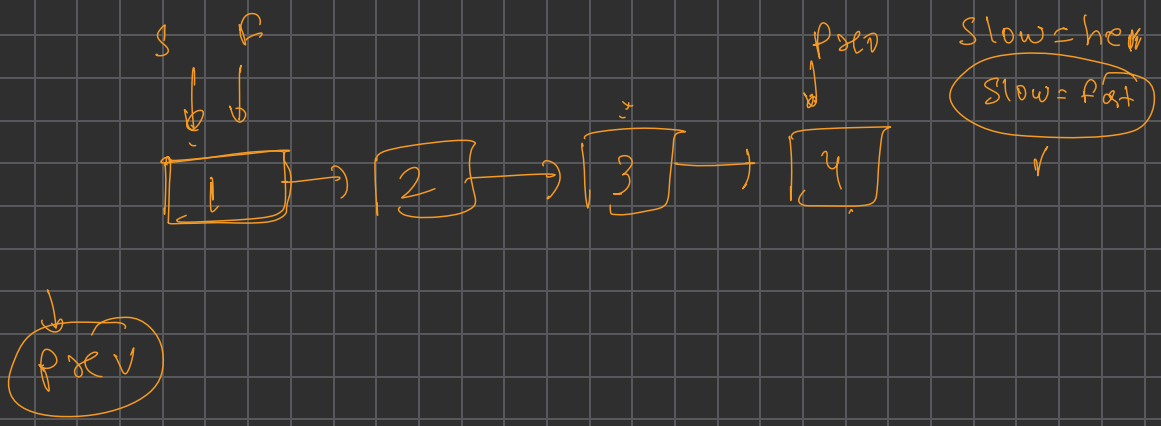
$\boxed{A = B + 2C}$

$A = C$

$A = (B+C) + C$

$A + B = 3B + 3C$

$A = 2(B+C) + C$


s
f
prev
slow = head
slow = fast
1
2
3
4
prev